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COMMENTARY

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Digital health services in Bangladesh—The need for a sustainable design framework

Arnob Zahid[1] | **Uday N. Yadav**[2] | **Sabuj K. Mistry**[3]

[1]Waikato Management School, University of Waikato, Hamilton, New Zealand

[2]ANU College of Health and Medicine, Australian National University, Canberra, Australia

[3]School of Population Health, University of New South Wales, Sydney, Australia

Correspondence

Arnob Zahid, Waikato Management School, University of Waikato, Hamilton, New Zealand.
Email: azahid@waikato.ac.nz

Abstract

Bangladesh received the United Nations Award for DigitalHealth Development in 2011. Inspired by the recognition, the country has launched several digital health initiatives over the last decade. However, despite achievements in expanding information and communication technologies, recent studies show that current digital health services are difficult to use, fail to meet user needs, are subject to threats in data privacy and security, lack a robust digital infrastructure, and do not benefit from a cohesive policy and regulatory environment. As the country currently deals with the post-COVID-19 consequences, it is high time for its healthcare policymakers to address these challenges and take the necessary actions to establish a people-centered framework for digital health service design and operations. In this commentary, we proposed a design framework for digital health services that address these ongoing challenges in the digital health services of Bangladesh. We believe the proposed framework stands to help Bangladesh and other coutnries attain sustainability.

KEYWORDS

digital health, patient-centric framework, policy design, sustainability

Key Points

- Most digital health services in Bangladesh lack usability, user-centricity, data privacy and security, digital infrastructure, technical and resource expertise, digital literacy, motivation, and trust of all healthcare stakeholders.
- It is time for Bangladesh's health policymakers to address the ongoing challenges that demand a

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people-centred digital health service design and operations framework.
- This policy perspective proposes a design framework for digital health services that addresses the ongoing challenges in the digital health services of Bangladesh.

INTRODUCTION

In 2009 Bangladesh set a target to become a fully digital nation by 2021, an effort known as "Digital Bangladesh" (Islam & Grönlund, 2011). Since then, the nation has made remarkable progress in expanding information and communication technologies (ICT). The country has already attained nationwide mobile network coverage (Statista, 2018), which includes 54% of mobile phones, 41% of smartphones ("The Daily Star," 2021), and 28.8% of internet users of the total population ("Dhaka Tribune," 2021). Furthermore, the country has taken various initiatives to increase financial resources to access ICT services and improve peoples' digital literacy (Rosa, 2017). By aligning with the "Digital Bangladesh" vision, the country has also made a massive technology deployment and transformation in the health sector over the last decade. This includes the development and implementation of District Health Information Software 2 (DHIS2), Digital Dashboard, Shared Health Record (SHR) system, National Health Data Warehouse, Digital Health Hub, Civil Registration and vital statistics (CRVS) system, Video Conferencing, Bulk Messaging Service, Telemedicine, Teleconsultation, E-prescription, and E-referral (Ahmed et al., 2014; Ahmed et al., 2014; Khan et al., 2019). However, these implementations only ensure the country's foundational readiness for digital health and lack in ensuring sustainability due to the impact of different ongoing challenges such as limitations in usability, user-centricity, data privacy and security, digital infrastructure, technical and resource expertise, budgetary constraints, and absence of regulatory and policy guideline in the existing context (Chowdhury et al., 2021; Gordon et al., 2020; Hoque et al., 2014; Islam, 2020; Khatun & Khanam Sima, 2015; Kiberu et al., 2017; Mistry, 2021). These challenges directly affect peoples' motivation and trust (Akter et al., 2012; Chowdhury et al., 2021; Grath-Lone et al., 2021; Sorwar et al., 2016). In addition, a few studies have examined localized digital health services (Ahmed et al., 2014; Ahsan & Raihan, 2013; Haque et al., 2012; Islam, 2020; Khan et al., 2019; Mistry, 2021). Most of these research studies address chronic diseases only (diabetes, cardiovascular diseases, cancer) (Islam, 2020), with a few exceptions such as maternal and child and mental and behavioral health (Mistry, 2021; World Health Organization [WHO, 2020]). However, the efficacy of these areas is yet to be fully understood (Islam, 2020).

The maturity of a patient-centric digital health service depends on its usage, resources and ability, interoperability, and impact on the operating health system (Flott et al., 2016). In this data-driven era, these digital health initiatives are unable to sustain without addressing those burning challenges. To be successful, Bangladesh needs to explore the possibility of integrating emerging information technologies within the existing infrastructure to address the current and future data needs of digital health, a data-centric, predictive (analytics-based), scalable digital health system (Svensson, 2019; Zahid et al., 2021). Furthermore, to achieve sustainability, Bangladesh must develop a practical system design framework (Islam, 2020; Sorwar et al., 2016) for digital health initiatives, both to establish standard guiding principles and ensure effective regulatory monitoring.

Bangladesh initiated a national digital health strategy in August 2019 (WHO, 2019a), though it has yet to be implemented (Bangladesh Ministry of Health and Family Welfare [MoHFW], 2020). As part of the development roadmap, the country recently participated in a four-day consultative workshop facilitated by the WHO, 2021. However, to the best of our

knowledge, the government has established no public collaborations and has not reportedly on what, if any, progress it has made with the program. Moreover, as the post-COVID-19 consequences continue in Bangladesh, progress is further delayed. Therefore, while the national digital health strategy development is in progress, we would like to draw the attention of the country's key health policymakers to consider the proposed system design framework in designing and developing digital health services for a sustainable digital health ecosystem.

SUSTAINABLE DESIGN FRAMEWORK FOR DIGITAL HEALTH SERVICES

Figure 1 depicts a system design framework that could serve as the basis of a national digital health strategy. The proposed framework delineates the technological aspects of sustainability in digital healthcare and the needed actions to achieve it in operating services. Furthermore, it addresses the ongoing practical challenges of usability, user adoption, data privacy and security, transparency, traceability, accountability and auditability, data exchange, scalability in digital health services, and the potential to improve the existing real-world context.

User-centricity

User centricity is vital for a system's success, especially in a progressive data-driven environment like Bangladesh. As most of the existing digital health initiatives of the country lack user-centricity (Alam & Khanam, 2023; Islam, 2020), policymakers must establish a user-centric system design guideline to ensure active users (e.g., patients, physicians, caregivers, and public health practitioners) engagement and derive success. Dashboards and UX design will follow from these multiple categories of users. To ensure user-centricity, digital health services should be intuitive and straightforward (Labrique et al., 2018; Nath & Sharp, 2015). The framework should also incorporate cultural aspects (values, traditions,



FIGURE 1 A conceptual design framework for Digital Health Service.

translation, environment) while designing services and interventions (Marwaha & Kvedar, 2021). Moreover, digital health services should complement users' practical, clinical, and emotional needs (Cresswell et al., 2013; Imison et al., 2016; Principles for Digital Development Working Group, 2016d; Stickdorn & Schneider, 2012; UK Department of Health & Social Care, 2019). Improved interoperability (discussed below) will help in the pivot toward user-centricity, making it easier to deploy locally customizable solutions based on a syntactically consistent data platform (WHO, 2019b). Digital health services may adopt an established health data documentation standard from a matured digital health entity such as HIMSS (Healthcare Information and Management Systems Society) or UKHDRA (UK Health Data Research Alliance) to support the system's interoperability need. In addition, it should be customizable, representing the users' lifestyle (Cheon et al., 2016). Communication between stakeholders through digital health services should be concise, exchanging minimal essential information (Jackson et al., 2018; Phillips et al., 2015). Furthermore, besides computers, digital health services should be accessible through smart gadgets such as smartphones, notebooks, and tablets, all of which enrich health equity in remote setups, bridging the digital divide. Lastly, digital health services should include an effective user feedback system to facilitate further design and development of user-centricity, advancing the efficacy of digital health services (Principles for Digital Development Working Group, 2016d).

Security and privacy

Due to the recent growth of cyberattacks (Global Digital Health Partnership [GDHP 2019b]) and the possibility of unauthorized access to digital health systems (Brothers & Rothstein, 2015), data security and privacy are critical components of digital health services. For Bangladesh, this is crucial as the regulation of data security and privacy policy lacks the protection of contemporary data ownership, access, control, sharing, storage, and anonymization needs (the latter of which is the irreversible removal of the link between an individual and his/her health record data to the degree that it would be virtually impossible to re-establish the link) (Alam, 2018; Chevrier et al., 2019; Hoque et al., 2014). Data security and privacy policies concerned with digital health services should explicitly define data sovereignty, ownership, access, storage, and control over data usage in simple language (Principles for Digital Development Working Group, 2016a). All personally identifiable information should be anonymized for any usage by the service users (Ipsos MORI, 2014; California Institute for Telecommunications and Information Technology (CITIT), 2014; Vithiatharan, 2014). Where an exception applies, personally identifiable information's deanonymization (a reverse engineering process where data de-identification is cross-referenced with other data sources to reidentify the personally identifiable information) must be strictly limited. It should be in accordance with the principles of limited purpose, data minimization, protection, and transparency (European Union [EU]., 2018). The services must incorporate data security by design and default (EU, 2018).

Besides conventional techniques, digital health services should also incorporate contemporary security techniques such as multifactor authentication (Borde, 2007; Mikkelsen et al., 2020) or data access, management, and control using robust solutions such as Distributed Ledger Technology (DLT, a digital system for recording the transaction in which the transactions and their details are recorded in multiple places simultaneously). Unlike traditional databases, distributed ledgers have no data storage center or administration functionality (Alladi et al., 2019; Nasajpour et al., 2020; Sharma et al., 2019; Troy & Mary, 2022; Zahid et al., 2022). Furthermore, digital health services should include appropriate mechanisms to circulate breach notifications (to the affected user),

system updates, security updates, or changes (Fernandez-Aleman et al., 2015; Mikkelsen et al., 2020). Including these in the data security and privacy policy will ensure transparency in service operations (EU, 2018), growing trust among users and encouraging them to engage actively with digital health services.

Traceability

Due to disruptive changes and a gradual increase in breach incidents, traceability has become a significant need for digital health services. In today's data-driven world, digital health services need to be equipped with digital tracking (Agarwal et al., 2018; Frøen et al., 2016; WHO, 2018). Digital tracking has the potential to: (a) minimize delays in treatment and care delivery, (b) assist health practitioners in decision-making at the point of care, (c) provide personalized healthcare in the care cycle, (d) schedule follow-up consultation, and other relevant services (e.g., diagnosis, pathological tests), (e) facilitate in maintaining checklists for case management, and (f) reduce time and cost of healthcare and delivery at remote settings (WHO, 2019b). Digital tracking is critical to a data-driven society, both to attract users and engage them in digital health services for care management and decision-making (The Office of the National Coordinator for Health Information Technology, 2018). WHO (2019b) recommends digital tracking in sharing health status and related services (e.g., appointment scheduling and payment). Tracking helps address population inequities and weak spot(s) in the health systems and services (e.g., when community workers went into the field to address COVID-19 outbreaks). It provides healthcare practitioners, patients, and public health professionals with the needed linkage(s) to “close the loop” of unmet healthcare needs. Overall, digital tracking ensures accountability at the individual and population health levels.

Digital tracking must also ensure strict data confidentiality. Digital health services should incorporate appropriate technological solutions (Encryption Schema, Artificial Intelligence, DLT) to ensure the required privacy and confidentiality for the system. They should establish a unique identity management system for all users (Imison et al., 2016; WHO, 2019b; Yao et al., 2010). The unique identification system is capable of: (a) assisting in patient searches; (b) reducing redundant entries in patient registries; and (c) improving intervention efficiency, treatment quality, and care delivery (WHO, 2019b). It has the potential to facilitate information exchange, or interoperability, across different healthcare providers, digital intervention, and population health (WHO, 2019b). Bangladesh has already transformed its national identification system and birth registry into a digital format (Election Commission Bangladesh, 2014). Digital health services should utilize these achievements in developing a unique health index for all its residents.

Transparency

In the digital world, 'transparency' means outcomes are predictable from the point of input in a system cycle (Rouse, 2014b). Transparency is one of the key demands users make of digital health services (Cordina & Greenberg, 2019). Transparency in digital health services can be developed by establishing a data flow map and can assist in classifying instances such as data processing information (including trace and track record) and interchanged data types (UK Department of Health & Social Care, 2019).

Furthermore, transparency can help digital health services to comply with regulatory standards. However, a data flow map may cause data misuse, such as niche advertising (Bigelow, 2018; Crawford et al., 2015). Therefore, the policymakers must prioritize this in

developing the digital health strategy and establishing a health data usage policy for third parties. Besides, digital health services may incorporate appropriate information technology such as Artificial Intelligence, Machine Learning, and Deep Learning, including details of the technology adopted (e.g., schema, mechanism, protocol, algorithm information) (Shneiderman, 2022). It potentially establishes an informed consent management mechanism within the system and ensures the best transparency practice for its users.

Accountability and auditability

Due to the gradual demand for transparency in digital health services (Cordina & Greenberg, 2019), accountability has become a significant issue in care services and delivery. Accountability supports improving a system's auditability. Auditability means examining and evaluating the existing system infrastructure, executions, and regulation policies (Rouse, 2014a). Accountability and auditability improve the transparency and traceability aspects of a digital health service. Therefore, to ensure systems' information transparency and traceability, digital health services should maintain standards (local or international) for security and performance evaluation. It may involve collaboration or include initiatives such as: (a) Computer Incident Response Centre (GDHP, 2019b; Ruefle et al., 2013), (b) Real-time threat sharing among different health stakeholders using traffic light protocol (GDHP, 2019b), (c) Common Vulnerability Scoring System (CVSS) for all medical devices used in the digital health (EU, 2018; Mell et al., 2007; Skierka, 2018), and (d) integrating Privacy Impact Assessment (PIA) (European Commission, 2016; EU, 2018). These mechanisms and initiatives ensure accountability and auditability of the services as per the regulatory policy (subject to availability and international standards) and add value to care delivery.

Interoperability

By definition, interoperability means the capability of strategic data exchange, access, integration, and use within health information systems, applications, and devices (HIMSS, 2020). It ensures the “standard meaning” of data for all the connected healthcare systems (Feldman et al., 2018). Interoperability enables seamless health data exchange within or beyond the organization and regional boundary. In the current digital health context, interoperability has become a precondition for digital health services. To achieve interoperability, digital health services must confidently confirm user identification in the system (GDHP, 2019b). It is achievable using standardized identifiers such as the National Health Index, a unique number assigned to each person using health and disability support services (Ministry of Health New Zealand, 2021). As Bangladesh has already established such an identification system (National Identification and Birth Registration Number), policymakers must act quickly to develop an NHI to enable interoperability.

Digital health services must adopt a common interoperability standard for their ecosystem (European Commission, 2014; Ferranti et al., 2006; GDHP, 2019b; Liaw et al., 2014; McMorrow, 2014). They may consider adapting open-source platforms (subject to budget or preference) such as OpenMRS (www.openmrs.org), CommCare (www.dimagi.com/commcare), OpenSRP (https://ona.io/), and OpenDataKit (https://opendatakit.org/). They may also consider proprietary standards such as FHIR (www.hl7.org/fhir/overview.html). These interoperability frameworks are easy to adapt and offer flexible feature support. Interoperability can potentially improve access equity, quality of coordinated care, and service delivery of digital health systems (GDHP, 2019b). However, the decision of how to implement interoperability may also have implications for transnational patients (Lee et al.,

2021). Policymakers, therefore, must consider interoperability for the ecosystem with significance.

Scalability

Scalability is the expandability of the system in proportion to the system's gradual needs (Bondi, 2000). User expect to drive it over time. As the current technology infrastructure of digital health services is becoming ineffectual in supporting the current and future health data needs (Svensson, 2019; Zahid et al., 2021), providers must take necessary actions for system scalability. First, policymakers must develop a standard guideline for facilitating emerging information technologies such as Big Data, Artificial Intelligence, Machine Learning, Deep Learning, Augmented technologies, Digital Twins, and Fitness devices into the system. In addition, they should strictly monitor the status of minimum data quality standards (GDHP, 2019a, 2019b; Principles for Digital Development Working Group, 2016b) in information exchange within the ecosystem. Major governing entities such as WHO (World Health Organization), IMF (International Monetary Fund), USAID (United States Agency for International Development), and United Nations have already introduced data quality assessment frameworks (MEASURE Evaluation, 2018; IMF, 2023) and strategoes (IMF, 2013; WHO, 2016), and fundamental principles (UN, 2015) that can be used as reference points for evaluation. Digital health service providers should also employ effective impact evaluation matrices (considering multiple factors such as priorities, expertise, time, and investment) for system scalability and risk (GDHP, 2019b; Principles for Digital Development Working Group, 2016b). These steps will ensure effective assessment outcomes for the system scale (Principles for Digital Development Working Group, 2016c). Digital health service providers should also consider effective collaboration and iteration in system design and development processes to ensure the highest possibility for system scale.

DISCUSSION

Sustainability in digital health services depends on effective system scalability and interoperability. To harness scalability and interoperability, digital health services need to ensure effective presence and support for user-centricity, data security and privacy, traceability, transparency, auditability, and accountability in the system. Digital health services should adopt agile design processes for their operations and service quality improvement. They should accommodate regular technological updates by integrating emerging information technologies (Zahid, 2021; Zahid et al., 2021). However, this updating practice sometimes involves large-scale technological implementation and may include risk of failure (as evident in developed economies). Therefore, integrating emerging technologies should be determined on a small scale, incorporating a low-tech approach (Sawadogo et al., 2021). In this way, local communities would be encouraged to innovate in agile ways with as much emphasis on how their goals could be supported by low-tech service provision as on links to digital infrastructure (Norman, 2023).

As Figure 1 depicts, realizing such a comprehensive framework requires technical resources, expertise, and consistent financial support (Labrique et al., 2018). The government of Bangladesh can arrange the needed funding in various formats (e.g., state fund, public–private partnership fund, international donors) with effective management, monitoring, and control support from the MoHFW. Figure 2 illustrates the strategies and emerging technologies that can potentially realize the much-needed framework and achieve the desired sustainability.

The sustainability of such a proposed framework includes environmental impact and energetical cost associated with ICTs (Chevance et al., 2020; Freitag et al., 2021; Waterman et al., 2015). Therefore, the government of Bangladesh should address these perspectives

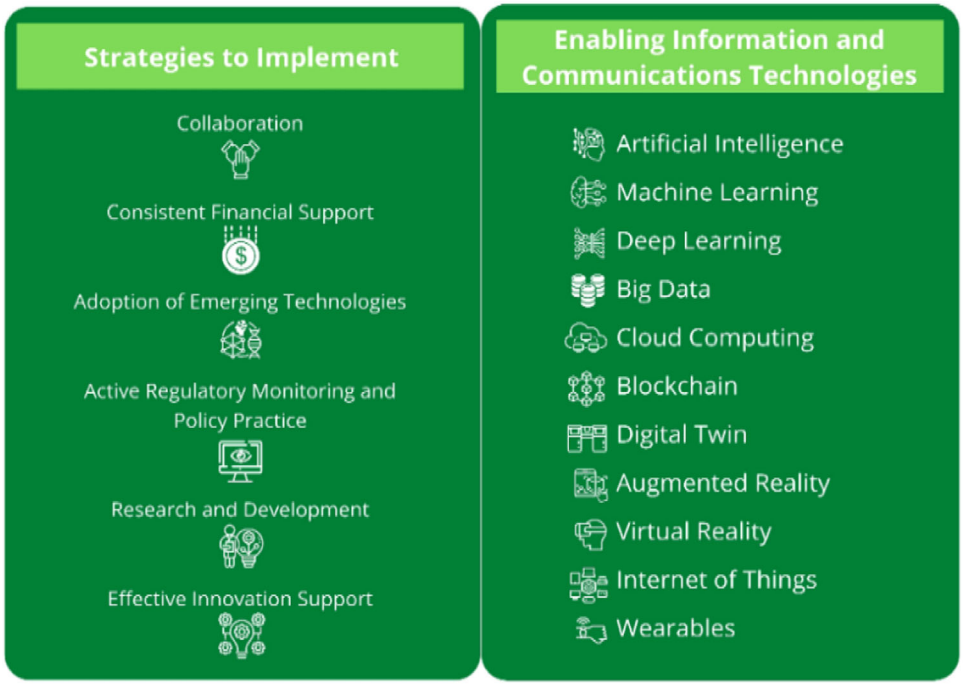


FIGURE 2 Potential implementation strategies and emerging technologies for sustainable digital health service.

with significance and determine appropriate actions with supportive arrangements. For instance, they may review the other available frameworks in neighboring countries (if any) or matured digital health systems (UK Department of Health & Social Care, 2019). Furthermore, they may evaluate and analyze Bangladesh's digital health literacy context and develop a precise action plan addressing the existing constraints.

The proposed framework highlights the needed technological implementations and actions to achieve sustainability in digital health services. It supports service innovation by facilitating the continuous iteration of system design and development. The framework is potential to address the ongoing challenges of usability, user adoption, data privacy and security, transparency, data exchange, and scalability in digital health services. We believe that considering such a framework in developing the national digital health strategy will potentially develop an effective citizen-centric sustainable digital health platform for the country. Lastly, we believe that adopting the proposed framework will build users' trust in the system, ensure active user participation (motivation) (Zahid & Sharma, 2023), and derive success for digital health services.

CONCLUSION

The Covid-19 pandemic made clear that information technologies play central roles in global healthcare (Buehler et al., 2020; Scott, 2020; Tonby et al., 2020). Healthcare stakeholders now have recognized the significance of data-driven digital health services in ensuring the sustainability of healthcare systems (Zahid & Sharma, 2023). By emphasizing the importance of system design principles, we have proposed a framework in this commentary for effectively implementing digital health services in Bangladesh. As the national digital

health strategy development is under-way, we anticipate that our proposed framework will actively support policymakers in formulating a comprehensive and data-centric approach. Doing so, Bangladesh can emerge as a beacon of digital healthcare progress and serve as a role model for other nations to emulate.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ETHICS STATEMENT

The article does not report primary data or raise issues related to informed consent and confidentiality. It did not require formal ethics committee review.

ORCID

Arnob Zahid http://orcid.org/0000-0001-6311-7271
Uday N. Yadav http://orcid.org/0000-0002-6626-1604
Sabuj K. Mistry http://orcid.org/0000-0001-6100-6076

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AUTHOR BIOGRAPHIES

Arnob Zahid, PhD, is a Digital Business lecturer at the Waikato Management School, University of Waikato, New Zealand. He is a design science researcher, experienced in studying qualitative social, behavioral, and economic interactions and different aspects of digital technologies, analytics, artificial intelligence, and Blockchain applications in the Personalized Digital Health spectrum. His research interest includes Digital Health Systems, Personalized Digital Healthcare Solutions, Information Systems Analysis, Design, Innovation, and Sustainability. He is an active member of renowned digital health and Information Systems societies, notably the Healthcare Information and Management Systems Society (HIMSS), Association for Information Systems (AIS), and Health Information New Zealand (HiNZ). He is a reviewer of multiple globally recognized journals, notably the Journal of Human-centric Computing and Information Sciences, Computer (IEEE Computer Society), Journal of Industrial Management and Data Systems, and International Journal of Information Management. He has been invited as a guest speaker to different global institutions to discuss the advancement of personalized digital healthcare solutions in the current data-driven context. His publications include globally recognized outlets such as Mayo Clinic Proceedings: Digital Health, International Journal of Medical Informatics, World Medical and Health Policy.

Uday N. Yadav, PhD, is an early career Implementation scientist, and his research interest include non-communicable diseases, global health, Indigenous health, primary health care, and evidence-based policy. He holds a Research Fellow (Implementation science) position at National Centre for Aboriginal and Torres Strait Islander Wellbeing Research, The Australian National University, Australia. Dr. Yadav's research has two major strands: assessing health problems and implementing co-designed solutions to address those problems. Dr. Yadav aims to address global health issues by implementing people-centred co-designed solutions and policy and advocacy with a special focus on unserved populations.

Sabuj K. Mistry, MPH, is a public health professional having more than seven years' experience in public health research and academics. His research interest lies within chronic disease management, integrated care, primary care, global health, and community health care. Mr. Sabuj is a Lecturer of International Public Health at the School of Population Health, UNSW Sydney, Australia. He holds an exceptional track record in public health research with more than 80 peer-reviewed publications (~1700 citations) that informed overten international policies.

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